Validity and Accuracy of Impulse-Response Models for Modeling and Predicting Training Effects on Performance of Swimmers

THIERRY BUSSO[1], and SÉBASTIEN CHALENCON[2]

[1]*Laboratoire Interuniversitaire de Biologie de la Motricité, Université Jean Monnet Saint-Etienne, Lyon 1, Université Savoie Mont-Blanc, Saint-Etienne, FRANCE; and* [2]*Club des Dauphins de Guilherand-Granges, Guilherand-Granges, FRANCE*

ABSTRACT

BUSSO, T., and S. CHALENCON. Validity and Accuracy of Impulse-Response Models for Modeling and Predicting Training Effects on Performance of Swimmers. *Med. Sci. Sports Exerc.*, Vol. 55, No. 7, pp. 1274–1285, 2023. **Purpose:** The aim of this study was to compare the suitability of models for practical applications in training planning. **Methods:** We tested six impulse-response models, including Banister's model (Model Ba), a variable dose–response model (Model Bu), and indirect-response models differing in the way they account or not for the effect of previous training on the ability to respond effectively to a given session. Data from 11 swimmers were collected during 61 wk across two competitive seasons. Daily training load was calculated from the number of pool-kilometers and dry land workout equivalents, weighted according to intensity. Performance was determined from 50-m trials done during training sessions twice a week. Models were ranked on the base of Aikaike's information criterion along with measures of goodness of fit. **Results:** Models Ba and Bu gave the greatest Akaike weights, 0.339 ± 0.254 and 0.360 ± 0.296, respectively. Their estimates were used to determine the evolution of performance over time after a training session and the optimal characteristics of taper. The data of the first 20 wk were used to train these two models and predict performance for the after 8 wk (validation data set 1) and for the following season (validation data set 2). The mean absolute percentage error between real and predicted performance using Model Ba was 2.02% ± 0.65% and 2.69% ± 1.23% for validation data sets 1 and 2, respectively, and 2.17% ± 0.65% and 2.56% ± 0.79% with Model Bu. **Conclusions:** The findings showed that although the two top-ranked models gave relevant approximations of the relationship between training and performance, their ability to predict future performance from past data was not satisfactory for individual training planning. **Key Words:** MODEL SELECTION, FATIGUE, TRAINING ADAPTATION, TAPERING

There is renewed interest in the mathematical models of training effects on performance to guide training planning (1–5). The most widely used models are impulse-response models, which were especially suited to acquiring knowledge on taper for performance peaking (6). However, progress toward implementing models in new applications for designing or monitoring individual athlete's training requires further exploration of model accuracy in predicting performance.

The use of impulse-response models in athletics was initiated by Banister and coworkers (7,8) to characterize the temporal effects of training on performance. Based on the mathematical formulation of the so-called *impulse response*, this model characterizes temporal variations in performance after a single training bout. For a given training intervention, the change in performance at a given day is obtained by summing over time the response to each single bout done the days before. Calculating performance for a period of intervention requires estimating model parameters by fitting the model to data on real performances. The data sets used to solve the model must include, on one hand, daily quantification of the amount of training, generally referred to as training loads (TL). These loads are expressed in an arbitrary unit (training unit in this study) depending on the metrics used to aggregate the different forms of exercise making up the training sessions. On the other hand, the data must also provide results of performance tests done regularly during the intervention so that the model can be

Address for correspondence: Thierry Busso, Ph.D., Laboratoire Interuniversitaire de Biologie de la Motricité, Université Jean Monnet Saint-Etienne, LIBM–Campus Santé Innovations–IRMIS, 10, Rue de la Marandière, 42270 Saint-Priest-en-Jarez, France; E-mail: busso@univ-st-etienne.fr.
Submitted for publication June 2022.
Accepted for publication January 2023.
Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's Web site (www.acsm-msse.org).

0195-9131/23/5507-1274/0
MEDICINE & SCIENCE IN SPORTS & EXERCISE®
Copyright © 2023 by the American College of Sports Medicine
DOI: 10.1249/MSS.0000000000003139

Copyright © 2023 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

parameterized and we can test its ability to describe the relationship between training and performance. It has been shown that measurement reliability and testing frequency (9) and methods for parameter estimation (10,11) can affect model accuracy. The number of athletes providing the data and the duration of data collection are also key to producing a powerful analysis.

Various impulse-response models have attempted to capture the main features of the relationship between variations of training effort and temporal variations of performance (7,8,12–14). In the original model of Banister et al. (7), Model Ba in this study, the impulse response of performance is the result of body adaptation (also referred to as fitness) and fatigue, which act positively and negatively on performance, respectively. Both positive and negative effects are modeled in an identical fashion using first-order kinetics (Fig. 1A and B). It results that performance decreases after training, requiring time, referred to as t_n, to recover to its initial level before a peak p_g is attained at the time t_g after which performance enhancement dissipates.

The model referred to as Model Bu in this study is an extension of Model Ba. It considers that the capacity to benefit from training is impaired by training overload and can be restored by reducing training (12). Contrary to Model Ba, Model Bu is based on the idea that the effects of a given training bout are dependent on previous training efforts. More precisely, the negative effect of a single session, i.e., an isolated bout, is lower than the effect of repeated similar bouts, i.e., a last bout effect. This brings on changes in t_n and t_g according to previous training (Fig. 1C and D). Model Bu outperformed Model Ba using data sets from volunteers in a training program designed for this purpose (12). However, when using data from elite athletes, the indicators of goodness of fit were in favor of Model Ba (15,16).

In Model Ba and Model Bu, the gain in adaptation is entirely produced immediately after the training session and

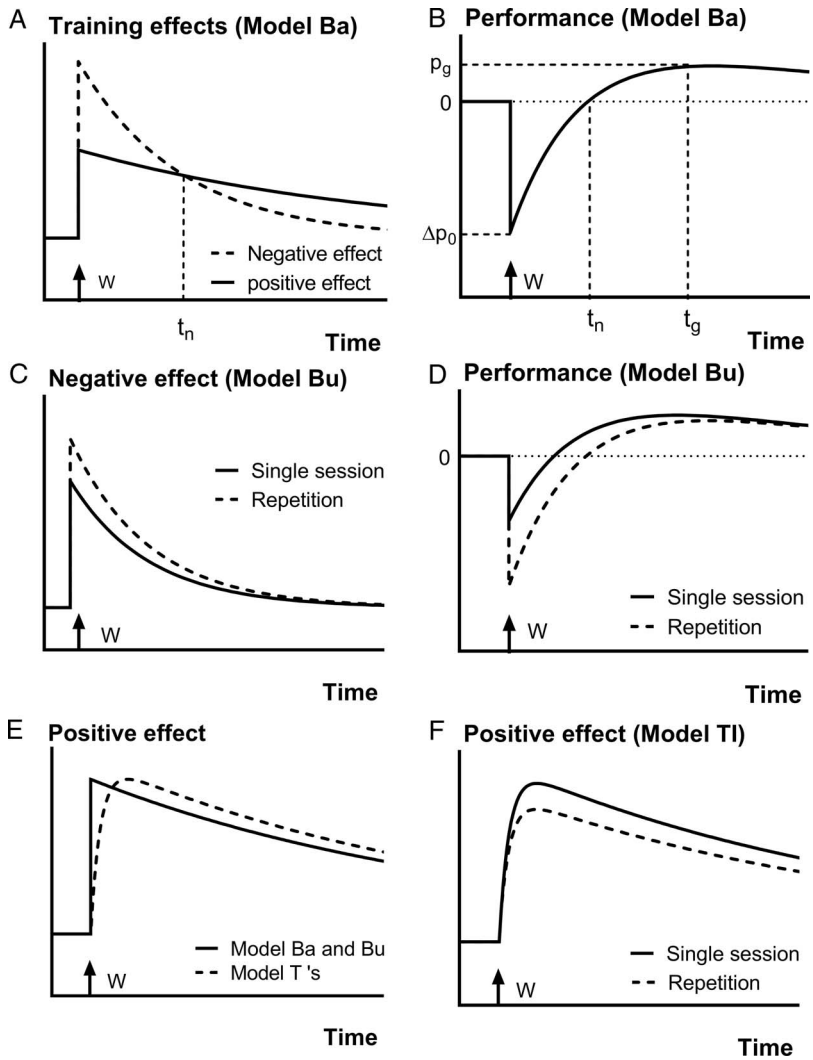

FIGURE 1—Schematic representation of impulse responses of models. A, Positive and negative effects of a single session, both being described by a single exponential with Model Ba. B, Performance over time based on the difference between positive and negative effects with Model Ba. ΔP0, decrease in performance after the training session; t_n, time needed to recover to the initial performance level; p_g, maximal gain in performance; t_g, time needed to reach p_g. C, Change in negative effect between response to an isolated training bout (single session) and to last bout after several repetitions (Model Bu). D, Change in performance over time between single session and repetition (Model Bu). E, Positive effect described by a single exponential (Models Ba and Bu) vs secondary signal model (Models TI, TF, TF2, and TIF). F, Positive effect by secondary signal model with inhibition due to previous sessions (Models TI and TIF).

Copyright © 2023 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

dissipates with a first-order rate constant. It is recognized, however, that adaptation results from physiological processes intervening during postexercise recovery. A better description of the delay in adaptation after training sessions is obtained by using double exponential to obtain a positive effect (8,14,15) or indirect-response model (13). The later was inspired by pharmacodynamics, considering the positive effect to be the result of the transformation of a signal secondary to the primary training stimulus (Fig. 1E). With indirect-response models, negative effects of training were better described by an inhibition of production of positive effects due to previous training sessions (Model TI; Fig. 1F) rather than fatigue with a first-order kinetics with or without inhibition process, Model TIF and Model TF, respectively. Nevertheless, it is likely that the absence of a fatigue component in the model makes it unsuitable for athletes whose performance capacity diminishes when their training is intensified. Furthermore, the inclusion of a fatigue factor, which changes kinetics according to previous training (Model TF2), was not tested in combination with an inhibition of the adaptation process.

Selecting a model as the best approximation among a set of models is a basis for statistical inference because it tells us what effects represented by which parameters are best supported by the data (17). Models differ in complexity, i.e., in the number of parameters to be identified by fitting the model to data. An underfitted model may not adequately capture the relationship between temporal variations in training and performance. Conversely, an overfitted model will tend to reproduce irrelevant variance in the training data, increasing the variability of estimates. The best model is ideally a parsimonious model able to capture the true relationship between the variables of interest while not overfitting the data. Often, R^2 adjusted and F-ratio tests are used to verify if an increase in model complexity is statistically relevant and would not lead to overfitting (12,15,16,18). However, Akaike's information criterion (AIC) is recognized as being a more powerful way to identify the best model because it ranks the models from best to worst by weighting the evidence that each model is likely the best (17,19).

Another way to test models is to measure their ability to predict future performance from past data used to parameterize or "train" the model. Both Model Ba and Model Bu were found to produce good predictions of performance in swimmers (20). This study used the data from a 15-wk training cycle for train the model (training data set) and those of the after 15-wk training cycle for testing the prediction (validation data set). The proximity in time between the two training cycles does question, however, to what point conclusions drawn for these observations can be generalized.

Our first aim was to generate large data sets from a group of swimmers with reliable and frequent performance measurements during an extended period to have substantial statistical power for estimating model parameters and testing the ability of the models to predict performance. Second, we aimed to rank a set of impulse-response models using AIC and their extensions along with indicators of goodness of fit. The estimates of the selected models were then used to explore the temporal variation of performance after a training session and the characteristics of the optimal taper period for comparison with data in the literature. Finally, the data from each athlete were split into training and validation data sets to determine the ability of the models to predict future performance.

METHODS

Data collection and processing. Data came from 11 swimmers, including 6 females specialized in 50-m or 50- and 100-m events (8 in freestyle and 3 in other strokes). They were 17.3 ± 2.9 yr old with competitive experience of 7.1 ± 2.9 yr at the beginning of this study. Best performances achieved during 50-m trials in 25-m pool were 75.9% ± 4.3% of world record speeds. All swimmers provided written informed consent before data use in accordance with procedures approved by Institutional Review Board of University Hospital of Saint-Etienne (reference number IRBM1262021/CHUSTE).

A single coach prescribed the training program for the entire group of swimmers and collected the data related to training efforts and performance testing from January to July 2019 (season 1) and then for 8 of them until March 2020 when training ceased because of COVID-19 lockdown (season 2). Season 1 covered the summer season comprising three training cycles ranging from weeks 1 to 28 before training was suspended for summer break. Season 2 covered the after winter season from weeks 34 to 48 (training cycle 4), then a break at Christmas before the beginning of the summer season from weeks 52 to 61(training cycle 5).

Swimming speed during a 1500-m time trial (S_{15}) was used to establish individual timetables of time per distance according to intensity levels: level 1 for active recovery (<90% of S_{15}), level 2 for basic endurance training (90%–94% of S_{15}), level 3 for aerobic training (95%–99% of S_{15}), level 4 for lactate threshold (100%–104% of S_{15}), level 5 for maximal oxygen uptake (105%–109% of S_{15}), level 6 for anaerobic capacity (110%–120% of S_{15}), and level 7 for sprints at maximal speed. Conditioning training comprised dry land exercises. Stroke rate and resistance settings were determined by the coach in correspondence with swimming speed, allowing him to design the training sessions according to intensity level. Training workouts were converted into equivalent swim distance at each intensity level.

TL was quantified each day from the weighted sum of the number of pool-kilometers swum and the dry land workout equivalent at each intensity level. Distances swum at zone 1 intensity were not included in the computation. The weighting factors for the remaining intensity zones were those proposed by Mujika et al. (21) as follows:

$$\mathrm{TL} = Z_2 + 2 \cdot Z_3 + 3 \cdot Z_4 + 5 \cdot Z_5 + 8\,(Z_6 + Z_7) \qquad [1]$$

where Z_2, Z_3, Z_4, Z_5, Z_6, and Z_7 are the number of kilometers swum at intensity levels 2, 3, 4, 5, 6, and 7, respectively. TL was set to 0 during periods of Christmas and summer breaks.

Copyright © 2023 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

Swimmers trained every day from Monday to Friday and rested the weekend except for participating in competitions participation. Training sessions of each Monday and Friday included a 50-m time trial in swimmer's stroke specialty in a 25-m pool. Each trial was supervised by a coach who blew a first whistle to position the swimmer on the starting block, a second whistle to give the start signal, and then manually timed the 50-m trial with a stopwatch (Seiko, Japan). Performance was assessed from the average speed during the trial expressed as a percentage of the first value of season 1. The reliability of this test assessed from the individual coefficient of variation from the first four trials of this study was 0.40% ± 0.23%.

Model computations. Six models were solved for each swimmer from daily TL and performance measured twice a week (see Supplemental Figure, Supplemental Digital Content 1, which gives a schematic representation of each model, http://links.lww.com/MSS/C802).

TL is considered as a discrete function, i.e., a series of impulses each day, w^i on day i, and the model performance on day i was estimated by mathematical recursion from the series of w before day i. The equations of each tested model were written as recursive sequences in which each term on a given day was defined as a function of the terms on either the same or the preceding day.

The models other than Model TI assume that performance is the result of change from baseline (p_{base}) because of cumulated adaptation to training (positive effect [pe]) and cumulated fatigue (negative effect [ne]).

$$\hat{p}^n = p_{\text{base}} + \text{pe}^n - \text{ne}^n \quad [2]$$

where pe^n, ne^n, and $\hat{p}^n$ are the estimations of pe, ne, and performance on day n, respectively, with pe^0, ne^0 initialized to 0.

Model Ba initiated by Banister et al. (7) is based on the idea that each training bout contributes to an increase in positive and negative effects proportional to corresponding w with multiplying factors (kin_{pe} and kin_{ne}, respectively, pe and ne), which decay away in the following days in an exponential fashion at separate rate constants (kout_{pe} and kout_{ne}, respectively). Model Ba is thus defined by the following equations:

$$\text{pe}^n = \sum_{i=1}^{n-1}\left(\text{kin}_{\text{pe}}.w^i + \text{pe}^{i-1}.e^{-\text{kout}_{\text{pe}}}\right) \quad [3]$$

$$\text{ne}^n = \sum_{i=1}^{n-1}\left(\text{kin}_{\text{ne}}.w^i + \text{ne}^{i-1}.e^{-\text{kout}_{\text{ne}}}\right) \quad [4]$$

Model Bu assumes that the multiplying factor for fatigue kin_{ne} is itself considered to increase by a quantity proportional to each w and decay exponentially away. The value of kin_{ne} at day i is estimated by mathematical recursion using a first-order filter with a gain terms $\text{kin}_{\text{in_ne}}$ and a rate constant $\text{kout}_{\text{in_ne}}$ as follows:

$$\text{kin}_{\text{ne}}{}^i = \sum_{j=1}^{i}\left(\text{kin}_{\text{in_ne}}.w^j + \text{kin}_{\text{ne}}{}^{j-1}.e^{-\text{kout}_{\text{in_ne}}}\right) \quad [5]$$

with $\text{kin}^0_{\text{in_ne}}$ initialized to 0.

Cumulated negative effect at day n is computed with Model Bu as follows:

$$\text{ne}^n = \sum_{i=1}^{n-1}\left(\text{kin}_{\text{ne}}{}^i.w^i + \text{ne}^{i-1}.e^{-\text{kout}_{\text{ne}}}\right) \quad [6]$$

Exponential decay of training effect is characterized by rate constant rather than time constant designated by τ, which is its reciprocal.

The other models tested in this study are based on an indirect response because pe results from a secondary signal produced by primary training stimulus. The secondary signal s accumulates with training and fades away exponentially as follows:

$$s^n = \sum_{i=1}^{n}\left(\text{kin}_s.w^i + s^{i-1}.e^{-\text{kout}_s}\right) \quad [7]$$

where kin_s is the multiplying factor for the secondary signal and kout_s is the rate constant with s^0 initialized to 0.

In Model TF and Model TF2, pe at a given day is assumed to increase by a quantity proportional to s the day before and decay with a first-order dynamic as follows:

$$\text{pe}^n = s^{n-1} + \text{pe}^{n-1}.e^{-\text{kout}_{\text{pe}}} \quad [8]$$

In Model TI and Model TIF, training acts negatively by inhibiting the secondary signal that drives the positive effect. The variable Inhib was added to describe this inhibition so that its value at day n is proportional to w^n as follows:

$$\text{Inhib}^n = k_{\text{inhib}} \cdot w^n \quad [9]$$

where k_{inhib} is the multiplying factor for inhibition process.

The production of pe varies in function of Inhib, resulting in the modification of equation 8 as follows:

$$pe^n = s^{n-1}.\left(1 - \text{Inhib}^{n-1}\right) + \text{pe}^{n-1}.e^{-\text{kout}_{\text{pe}}} \quad [10]$$

Performance with Model TI is p_{base} added to pe estimated by equation 9, i.e., without fatigue. In Model TF and Model TF2, pe is estimated from equation 7, i.e., without an inhibition process, but counterbalanced by fatigue estimated from ne given by equation 4 as in Model Ba (Model TF) or by equations 5 and 6 as in Model Bu (Model TF2). Finally, performance with Model TIF is estimated from the balance between pe with inhibition process given by equations 7 to 10 and ne given by equation 4.

Model fitting. The model parameters were determined by minimizing the residual sum of square (RSS) between estimated and measured performance as follows:

$$\text{RSS} = \sum_{i=1}^{n}\left(p^i - \hat{p}^i\right)^2 \quad [11]$$

where n is the number of performance tests used to fit the model. RSS was minimized by using a hybrid method that combines standard linear regression with an L-BFGS-B algorithm, which is a quasi-Newton method allowing us to set lower and upper bounds on the estimates. The latter was implemented using constrOptim function in R package (22) for

Copyright © 2023 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

fitting the rate constants ($kout_{pe}$, $kout_{ne}$, $kout_{in_ne}$, and $kout_s$ according to the model) and k_{inhib} for Models TI and TIF. At each step of the optimization, the other parameters (i.e., the multiplying factor kin_{pe}, kin_{ne}, and kin_{in_ne} according to the model) were obtained by linear regression directly. This hybrid method allowed us to define the optimization starting point of and the search region for only a part of the entire set of model parameters. We verified that the solution given by this optimization method was not sensitive to the starting point of the computation. The details of the minimization procedure can be found in the supplementary information (see document, Supplemental Digital Content 2, which gives R script of the models, http://links.lww.com/MSS/C803).

To rank the models according to goodness of fit and AIC results, they were fitted to all the data collected for each swimmer giving n = 56 for 3 swimmers (season 1 only) and n = 104 for 8 swimmers (seasons 1 and 2). For testing their ability to predict performance from past data, models were fitted to the data of training cycles 1 and 2 (training data sets with n = 40 for each swimmer). The estimates for Model Ba and Model Bu were used to predict performance during cycle 3 for 11 swimmers and during cycles 4 and 5 for 8 swimmers.

Model estimates. In addition to model parameters, variables were calculated to characterize the time response of performance to a single training bout. t_n, the time to recover performance, and t_g, the time to peak performance after training completion, were computed from parameters of Model Ba as follows:

$$t_n = \frac{\ln\left(\mathrm{kin_{ne}}/\mathrm{kin_{pe}}\right)}{\mathrm{kout_{ne}} - \mathrm{kout_{pe}}} \quad [12]$$

and

$$t_g = \frac{\ln\left(\mathrm{kin_{ne}}\cdot\mathrm{kout_{ne}}/\mathrm{kin_{pe}}\cdot\mathrm{kout_{pe}}\right)}{\mathrm{kout_{ne}} - \mathrm{kout_{pe}}} \quad [13]$$

p_g, the maximal gain in performance after a training session at a given TL, is given by

$$p_g = \mathrm{kin_{pe}}\cdot\mathrm{TL}.e^{-\mathrm{kout_{pe}}\cdot t_g} - \mathrm{kin_{ne}}\cdot\mathrm{TL}\cdot e^{-\mathrm{kout_{ne}}\cdot t_g} \quad [14]$$

Calculation of these variables for Model Bu must account for variations in kin_{ne} according to cumulated training. Its lowest value is that for a single training bout dependent on corresponding TL (single session). If the same training session is repeated every day, kin_{ne} will tend toward a limit value dependent on both TL and $kout_{in_e}$ (asymptote). Both estimates were estimated as follows:

$$\mathrm{kin}_{ne}(\text{single session}) = \mathrm{kin_{ne}}\cdot\mathrm{TL} \quad [15]$$

and

$$\mathrm{kin}_{ne}(\text{asymptote}) = \frac{\mathrm{kin_{ne}}\cdot\mathrm{TL}}{1 - e^{-\mathrm{kout_{in_ne}}}} \quad [16]$$

Statistics. Let p be the number of parameters of each model: 5 for Models Ba and TI, 6 for Models Bu and TF, and 7 for Models TF2 and TIF. We calculated the following measures of goodness of fit (19): the coefficient of determination (R^2)

$$R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}} \quad [17]$$

where TSS is the total sum of squares. The mean square error (SE) was computed as the square root of RSS/($n - p$). F-ratio was computed to assess the statistical significance of one model fit with degrees of freedom (DF) set to $p - 1$ for the model and $n - p - 1$ for the residuals:

- the mean absolute percentage error (MAPE):

$$\mathrm{MAPE} = \sum_{i=1}^{n} \frac{\left|p^i - \hat{p}^i\right|}{\hat{p}^i}.100 \quad [18]$$

- the adjusted coefficient of determination (Adj.R^2):

$$\mathrm{Adj}.R^2 = 1 - \frac{n-1}{n-p}.\left(1 - R^2\right) \quad [19]$$

- the bias-corrected Akaike information criterion (AICc):

$$\mathrm{AICc} = 2\cdot p - 2\cdot \ln(L) + \frac{2p(p+1)}{n-p-1} \quad [20]$$

where ln(L) is the log-likelihood of the estimated model computed as follows:

$$\ln(L) = -0.5n\left[\ln(2\pi) + 1 - \ln(\mathrm{n}) + \ln\left(\sum_{i=1}^{n}\left(p^i - \hat{p}^i\right)^2\right)\right] \quad [21]$$

- the Akaike weights, w_i(AICc), provide weight-of-evidence measurements for each of the six models tested in this study:

$$w_i(\mathrm{AICc}) = \frac{\exp\left(-0.5.\Delta_i(\mathrm{AICc})\right)}{\sum_{k=1}^{6}\exp\left(-0.5.\Delta_k(\mathrm{AICc})\right)} \quad [22]$$

where i is the model number and Δi (AICc) is the difference between AICc of model i and the lowest AICc.

One-way repeated-measures ANOVA was conducted to compare between models of measures of goodness of fit and characteristics of impulse response. The reliability of the prediction of performances for the testing data sets was assessed from the calculation of systematic error, typical error, and MAPE between predicted and measured performances. Two-way mixed-effect ANOVA was done to compare between models and data sets. All analyses included Greenhouse–Geisser correction to ensure sphericity assumption. When significant differences were found, Tukey's *post hoc* test was used for pairwise comparisons. Paired t-test was conducted for other

Copyright © 2023 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

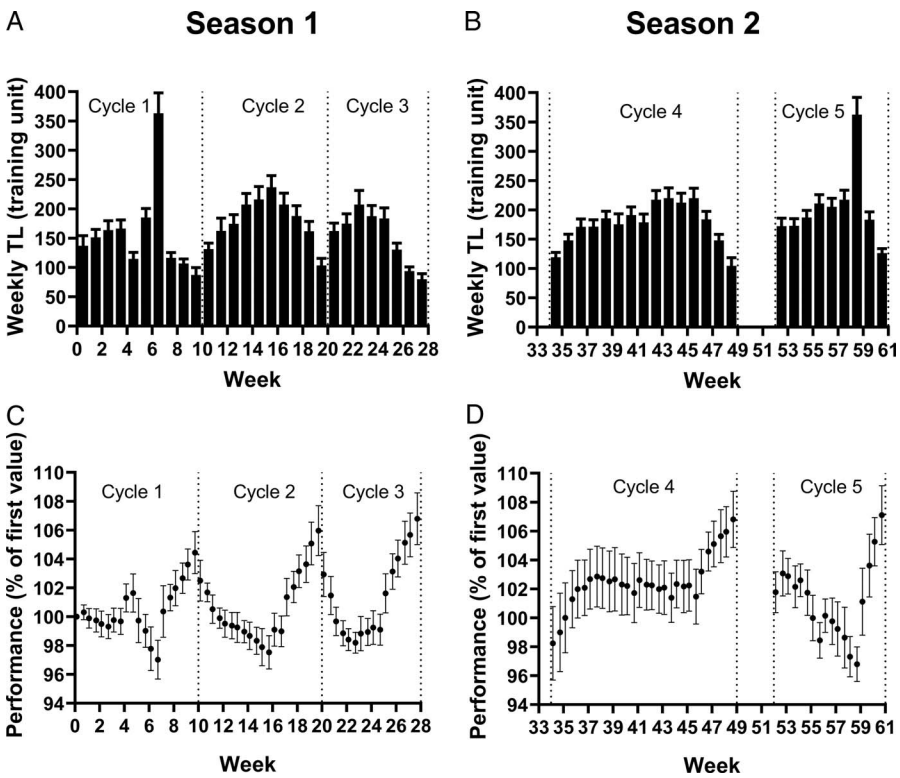

FIGURE 2—Distribution of TL and change in performance. A and B, Mean ± SD of weekly TL during seasons 1 and 2, respectively. C and D, Mean ± SD of performance relative to first value during seasons 1 and 2, respectively.

comparisons when appropriate. The significance level was set at $P < 0.05$.

RESULTS

Data sets. Figure 2A and B gives weekly TL for season 1 and season 2, respectively. Weekly TL were higher during the first part of each training cycle with a reduction during the 2 or 3 last weeks. The performance was maintained or decreased during the first part of each training cycle and then increased when TL was reduced until a peak before the beginning of the following cycle (Fig. 2C and D for seasons 1 and 2, respectively).

Model fits. Table 1 summarizes the indicators of goodness of fit of each model fitted to the data of the two seasons of training in each swimmer. R^2, Adj.R^2, and SE from Model TI were significantly lower than from other tested models ($P < 0.05$). F-ratio showed that Model TI only significantly fitted the performance of seven swimmers ($P < 0.05$ in 2 and $P < 0.001$ in 5), whereas other models significantly fitted all of swimmers individually ($P < 0.001$ for all data sets with the exception of 1 with Model TF and Model TIF, which was at $P < 0.05$).

No difference in R^2, Adj.R^2, and SE was observed between the models other than for TI. We used w(AIC), which includes model complexity (i.e., the number of model parameters), to rank the models. Individual w(AIC) scores were best in five swimmers with Model Ba and in six swimmers with Model Bu. No difference was observed at the group level in w(AIC) between Models Ba and Bu (Table 1). Based on these results, Model Ba and Model Bu rank equally as the best models among those tested in this study.

TABLE 1. Summary of goodness-of-fit and AIC results for models with all data of seasons 1 and 2.

Model	R^2	Adj.R^2	SE	MAPE (%)	AICc	w(AIC)
Ba (p = 5)	0.286 ± 0.100	0.250 ± 0.097	2.28 ± 0.53	1.82 ± 0.45	409 ± 112	0.339 ± 0.254
Bu (p = 6)	0.297 ± 0.073	0.252 ± 0.071	2.28 ± 0.52	1.80 ± 0.42	409 ± 109	0.360 ± 0.296
TI (p = 5)	0.135 ± 0.053*†	0.091 ± 0.062*†	2.52 ± 0.56*†	2.00 ± 0.44*†	425 ± 108*†	0.001 ± 0.002*†
TF (p = 6)	0.288 ± 0.098‡	0.243 ± 0.094*‡	2.29 ± 0.53*‡	1.81 ± 0.45‡	411 ± 112*‡	0.115 ± 0.082*‡
TF2 (p = 7)	0.304 ± 0.081‡	0.251 ± 0.077‡	2.29 ± 0.53‡	1.79 ± 0.43‡	411 ± 110†‡	0.123 ± 0.088‡
TIF (p = 7)	0.299 ± 0.097*‡§	0.246 ± 0.091‡	2.29 ± 0.53‡	1.80 ± 0.45*‡§	412 ± 112‡	0.061 ± 0.036*‡

Values are presented as mean ± SD. Number of performance points is 56 for 3 swimmers (season 1) and 104 for 8 swimmers (seasons 1 and 2).
*Significant difference ($P < 0.05$) from Model Ba.
†Significant difference ($P < 0.05$) from Model Bu.
‡Significant difference ($P < 0.05$) from Model TI.
§Significant difference ($P < 0.05$) from Model TF.
p, number of model parameters.

Copyright © 2023 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

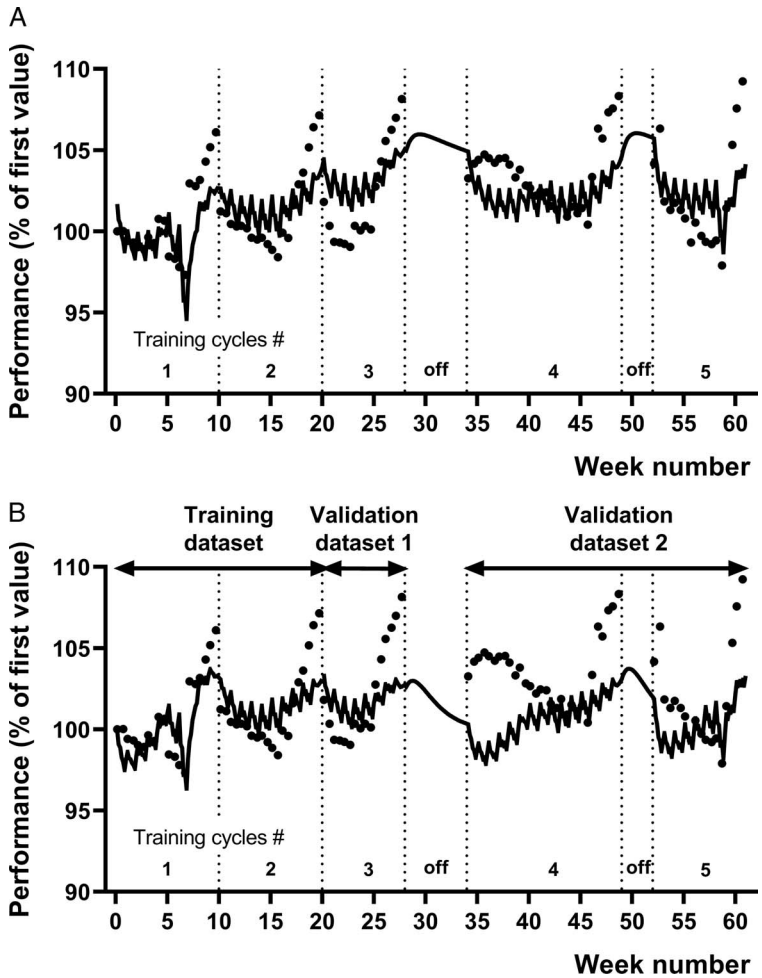

FIGURE 3—Modeling and prediction of performance for one representative swimmer with Model Ba. A, Model fit to all the data of season 1 and season 2. B, Model fit to the data of cycles 1 and 2 (training data set) and prediction tested using data of cycles 3, 4, and 5 (testing data sets). *Solid circle*: observed performance. *Continuous line*: modeled performance.

Figure 3A shows the performance fit with Model Ba from the entire data set of one representative swimmer. It shows substantial overestimation of the lowest values and underestimation of peak values at the end of each training cycle. This tendency is observable for every swimmer using both Model Ba and Model Bu. At the group level, the difference between Model Ba and real performance was different from zero ($P < 0.001$) at the end of each training cycle: −1.5% ± 0.9%, −3.3% ± 1.2%, −3.8% ± 1.2%, −3.9% ± 1.2%, and −4.1% ± 1.3% at the end of cycles 1 to 5, respectively. Estimates for Model Bu also significantly underestimated peak performance ($P < 0.001$): −1.9% ± 1.2%, −3.2% ± 1.4%, −4.0% ± 1.3%, −3.8% ± 1.2%, and −4.0% ± 1.2% at the end of cycles 1 to 5, respectively. Residuals and real performance were highly correlated in each swimmer: 0.84 ± 0.06 and 0.84 ± 0.04 for Model Ba and Model Bu, respectively.

Characteristics of performance over time after a training session (Fig. 4) were assessed from the parameters of Model Ba and Model Bu (see Supplemental Table, Supplemental Digital Content 3, which gives the estimates, http://links.lww.com/MSS/C804). Additionally, computer simulations were done to determine the duration and the rate of training reduction during taper, which would give the highest performance according to the form of the taper, i.e., with step, linear, or exponential training reduction, following previous reports (23–25). We computed optimal characteristics for a taper after regular training consisting in daily training equal to median TL (24.5 ± 2.5 training units) assumed to be long enough to stabilize performance and followed or not by a 20% step increase in training for 28 d. The characteristics of the optimal taper according to the form of training reduction are given in Table 2.

Validation data sets. Figure 3B shows prediction performance for Model Ba for one representative swimmer. The range of predicted values was narrower than that of real values with substantial underestimation and overestimation of the highest and the lowest performances, respectively. Table 3 gives the systematic and typical errors for the group of swimmers. For the first validation data sets, typical error was greater than with the training data set, and without significant systematic error. For the second testing data sets, each model led to significant systematic error and a typical error greater than with training data sets. MAPE values between predicted and real performance for the first data set were 2.02% ± 0.65% and 2.17% ± 0.65% for Model Ba and Model Bu, respectively. Greater MAPE values

Copyright © 2023 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

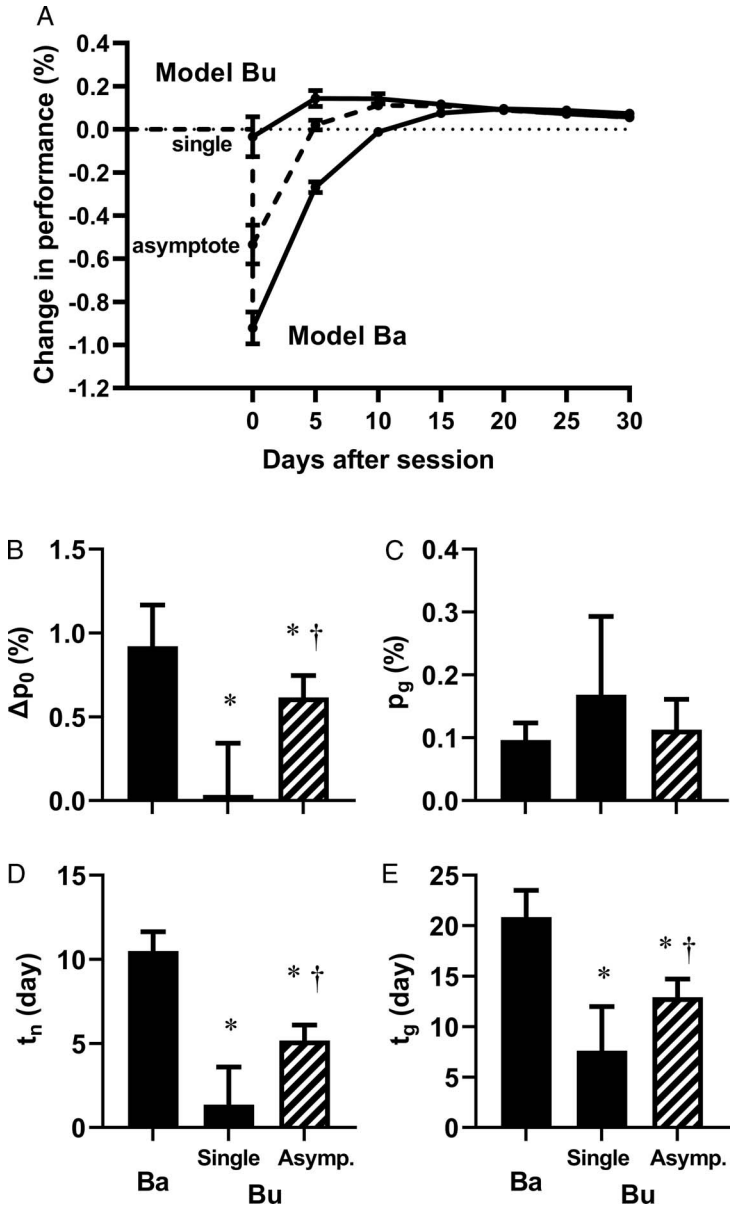

FIGURE 4—Mean ± SD of characteristics of impulse response to median TL from estimates of Models Ba and Bu. A, Change in performance after one session. B, ΔP0, initial decrease in performance. C, p_g, maximal gain in performance. D, t_n, time needed to recover to initial performance level. E, t_g: time needed to reach p_g. For Model Bu, results show the response to an isolated training bout (single session) and to the last bout at steady state after daily repetitions (asymptote). *Significantly different from Model Ba. †Significantly different from Model Bu for single session ($P < 0.05$).

were observed for the second testing data set, 2.69% ± 1.23% and 2.56% ± 0.79% for Model Ba and Model Bu, respectively.

DISCUSSION

The present findings are that Model Ba and Model Bu were relevant for modeling the change in performance with training in athletes and showed a somewhat lesser ability to predict individual future performance from past data in a given swimmer. Impulse-response models suffer, however, from some limitations indicated in a series of recent studies (3,9–11).

Model parameter estimation requires a great number of data points from valid, reliable, and frequent tests of performance. We paid great attention to meeting these key requirements, leading us to use a test mimicking competitive racing conditions twice a week over an extended period. The results of the four trials done the first 2 wk of this study showed a high reliability of this test. The coefficient of variation of 0.40% ± 0.23% was close to the value around 0.9% observed for field tests of sprint running (26). One limitation is the existence of local minima, resulting in convergence to a solution sensitive to the initial parameters chosen for the starting point of the algorithm (10,11). In our opinion, this troublesome problem is particularly relevant when all the parameters are estimated together using a typical fitting process. Finding initial estimates for the multiplying factors, i.e., kin_{pe}, kin_{ne}, or kin_{in_ne} depending on the model, is an awkward task because their estimates are dependent on the metrics used to quantify training and

Copyright © 2023 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

TABLE 2. Mean ± SD of estimated characteristics of the optimal simulated taper according to the form of the training reduction after regular training at the median TL and overload training with a step increase of 20% for 28 d.

	Form of the Training Reduction during Taper		
	Step	Linear	Exponential
Mean reduction (%)			
Regular training	86.9 ± 16.0	73.5 ± 12.3†	61.5 ± 12.6†
Regular training +20%	86.3 ± 13.8	73.3 ± 8.8†	66.4 ± 12.6†
Duration (d)			
Regular training	6.5 ± 1.2	9.0 ± 1.1†	8.2 ± 0.6†
Regular training +20%	7.6 ± 0.7*	9.5 ± 1.7*†	8.7 ± 1.0*†
Highest performance (% of pretraining)			
Regular training	103.24 ± 1.12	103.21 ± 1.13†	103.21 ± 1.13†
Regular training +20%	103.85 ± 1.30*	103.82 ± 1.31*†	103.83 ± 1.30*†

*Statistically different from regular training previous taper ($P < 0.05$).
†Statistically different from step taper ($P < 0.05$).

performance. This is the reason why we chose a hybrid approach to searching for parameter estimates, after checking that the solution identified was not sensitive to the starting point of the search for any of the models tested with our data. This hybrid approach used a quasi-Newton algorithm, but only for a part of the parameters, whereas the other parameters, i.e., kin_{pe}, kin_{ne}, or kin_{in_ne}, were estimated from simple regression at each step of the searching process.

Another important limitation is that it is possible to find different solutions that give close goodness-of-fit scores. This could preclude the use of model parameters as characteristics of one individual's response to training. We speculate that this might be due to compensation between model parameters, which when combined together could give similar performance impulse responses. To support this assertion, we searched for other solutions than those used in the main study. We have repeated all the analyses with more restrictive constraints for $kout_{pe}$ to find an alternative solution with much lower estimates for $kout_{pe}$. This resulted in large variations of the estimates of other parameters for each model (see document, Supplemental Digital Content 3, which gives details on these alternative solutions and their influence on model estimates, http://links.lww.com/MSS/C804). This alternative solution gave an optimized RSS close to that of the main studies. For Model Ba and Model Bu, the alternative sets of parameters gave small difference for the characteristics of the impulse response, although statistically significant differences were observed. As a consequence, results on optimal taper and prediction accuracy of performance are not significantly affected by this alternative solution. Because of the uncertainties of each model parameter, taken individually, it would be preferable to consider the impulse response obtained for an entire set of parameters when characterizing the response of one individual to training. In any case, this drawback regarding the estimation of model parameters did not appeared to significantly affect the findings of the present study.

The impulse-response models tested in this study differ in 1) the description of positive and negative effects of a single training bout and 2) their consideration, or not, of change in impulse response due to repetition of training bouts. One interest of indirect-response models lies in the differentiation of cumulated fatigue acting directly on performance and altering the athlete's capacity to adapt to the training. This can be particularly relevant when analyzing athlete behavior during overtraining (27). Indirect-response models have only been tested in participants in a controlled experiment who were not experienced athletes (13). In this his earlier study, Model TI outperformed other models. In the present data from swimmers, the TI model failed to effectively describe their variations in performance and should be thus discarded for data analysis in athletes. When negative effects of training (i.e., fatigue) are considered, indirect-response models fitted the data better than Model TI but did not better than Model Ba or Model Bu. AIC showed low weight of evidence in favor of Models TF, TF2, and TIF. Because of overfitting resulting in uncertainty in their estimates, these indirect-response models with fatigue components should be also discarded.

The absence of a clear difference in performance between Model Ba and Model Bu is in accordance with previous findings in athletes. On one hand, Model Bu was found to significantly improve performance fit compared with Model Ba in 7 out of 10 swimmers during a training period of 15 wk (20). On the other hand, a study in three elite swimmers found higher R^2 and lower AIC for Model Ba, compared with Model Bu (16). The authors pointed out that the frequency of performance tests could explain these discrepancies. They dispose of a limited number of performances in their data sets because they used results from competitions during a period of 400 d. The results of the present study also support this hypothesis when using only the first 20 wk, i.e., using only 40 values for performance. In this case, Adj.R^2 was significantly greater for Model Ba than Model Bu, 0.36 ± 0.09 versus 0.30 ± 0.09, respectively. In our previous report (20) and in the present study,

TABLE 3. Reliability of the performance predictions.

Model	Data Set	Systematic Error (%)	Typical Error (%)	MAPE (%)
Ba	Training	0	1.68 ± 0.45	1.34 ± 0.37
	Validation 1	0.16 ± 0.40	2.40 ± 0.74*	2.02 ± 0.65*
	Validation 2	−1.74 ± 1.23*†	2.43 ± 0.51*	2.69 ± 1.23*†
Bu	Training	0	1.73 ± 0.47	1.39 ± 0.39
	Validation 1	0.21 ± 0.42	2.56 ± 0.80*,‡	2.17 ± 0.70*
	Validation 2	−1.64 ± 1.15*†	2.32 ± 0.49*,‡	2.56 ± 0.79*†

Values are presented as mean ± SD. Systematic and typical errors are expressed as a percentage of first performance of season 1.
*Significant difference ($P < 0.05$) from Training data set.
†Significant difference ($P < 0.05$) from first validation data set.
‡Significant difference ($P < 0.05$) from Model Ba.

APPLIED SCIENCES

Copyright © 2023 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

we used performance tests done routinely during training sessions once and twice a week, respectively. Results of competitions are the gold standard to determine athlete performance, but the low number of competitions limits the statistical power when comparing modeling methods. This is why, despite possible bias, trials frequently repeated during the training sessions are an alternative that enables the collection of large data sets on which to train and test models. Because the number of performance measurements was a major limit, it would be preferable to use Model Ba for data sets smaller than that of the present study because it is the most parsimonious model.

Frequency of performance trials higher than in previous reports provided a larger amount of information, which could be fruitful for exploring models more complex than Model Ba. In return, greater numbers of data points could affect fit quality. Although R^2 values reported in the literature (28) ranged from 0.29 to 0.97, R^2 values for Model Ba and Model Bu obtained with the present data set were in the lower part of this range. The difference between model values and real performances showed that fit error was not randomly distributed. It is likely that the lower fit quality with a greater number of points was more due to greater loss of information than to greater noise. This suggests that Model Ba and Model Bu are not optimal models. Further work is necessary to find models that can better approximate the present large data sets, in the sense that models that loose less may serve to gain new insights into performance over time in adaptation to training.

Models could be used to probe the responses to solicitations and compare the effect of different training strategies (29,30). Estimates of t_n and t_g have been considered to be key values for optimizing tapering period before a competition (31). According to Model Ba, taper duration should be ranged between t_n and t_g because negative effect (fatigue) dissipates more quickly than positive effect (adaptation or fitness) (31), i.e., between 11.0 ± 1.9 and 21.2 ± 2.7 d, respectively, in the present study. The major flaw in this reasoning is that the best performance would be indeed achieved without training during at least t_n days. It is recognized that several weeks without training would be prejudicial for performance because of detraining. Furthermore, the large estimates of t_n, which is in theory the time needed to recover performance after any training session, are not supported by the temporal variation of several physiological parameters altered by exercise (32). Contrary to Model Ba, Model Bu attempts to account for change in the response to a given workout among cumulated training sessions. A designated session could be more difficult to cope with when training is intensified. Conversely, the tolerance to training is recovered when training is reduced. This is illustrated by Figure 4, which shows single bout and last bout effects at the asymptote calculated from the estimates of Model Bu. Mean recovery time for median TL would be ranged between 1 and 6 d depending on previous training. Compared with Model Ba, these estimates from Model Bu are in better agreement with within-subject variability in the time required to restore physiological alterations after working out (32). Additionally, computer simulations using estimates of Model Bu were done to explore the duration, form, and extent of training reduction to maximize performance during the taper period (23–25,33,34). Individual responses to training were simulated for accustomed training, evaluated as the median of TL, and for unaccustomed training as a 20% step increase in TL for 28 d. Contrary to Model Ba, estimates of Model Bu gave an optimal taper for around 1 wk with a training reduction of 80%–90% for a step taper and 60%–70% for a progressive taper. This matches with an earlier review of the literature, which suggests a training reduction of 50%–90% over 4–35 d regardless the form of the training reduction (35). It agrees less with a meta-analysis study, which showed that the most efficient taper for maximizing performance in swimmers was 1 or 2 wk with a progressive reduction of training volume by 40%–60%, without any modification of intensity (36). Contrary to our previous report in elite swimmers (23), greater solicitations before the taper require only a small increase in taper duration, which remains around 1 wk, and progressive reduction of training did not give better results than step reduction. These differences could be explained by the lower level of performance of the participants in the present study. Although our previous report in best nationally ranked swimmers (23) concluded that optimal taper would be a training reduction of 40% to 70% during 2 to 3 wk, depending on previous training, the present findings suggest for subelite swimmers shorter taper lasting around 1 wk with greater training reduction, around 60% to 90%.

In our analysis, we examined how well models predicted the future from past data and compared model performance over different time spans. The theoretical studies on taper characteristics based on model predictions (23–25,33,34) have to be distinguished from using one model to monitor training of a given individual because this kind of application would require a more precise assessment of the statistical confidence of the model prediction (37). Prediction accuracy was lower than in our previous report in swimmers (20). It could be explained by the closeness of the training cycles used to train the models and to test their prediction. It is the reason why we shared the data to generate two validation data sets per swimmer: first one for testing the prediction during training cycle immediately after the training period used to parameterize the models and a second one for testing the prediction for the subsequent season after the summer off-season. This design showed that the agreement in performance prediction dropped for the last cycle of season 1 to the two cycles of season 2 due to a large bias for season 2. Typical errors indicated Model Ba performed better for season 1, whereas Model Bu was better for season 2 but the differences in the performance of the two models were too small to be considered to be significant. Altogether, relative accuracy for predicting performance was around 2%–3% for both models, whereas the fluctuations of actual performance throughout the two seasons were between 3% below and 7% above the first value of season 1 (Fig. 2). This accuracy level does not effectively meet the need of athletes if models were to be implemented in practical application to assist coaches when planning training of a given

Copyright © 2023 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

individual. More accurate prediction is required before considering these models for such applications.

The literature provides other model structure solutions, which should also be tested for their ability to fit the observed performances and ultimately to predict future performance. One important feature not included in Model Ba or in Model Bu is that positive effects are linearly dependent on TL. This is a critical issue when attempting to better represent maladaptation because of overtraining. It is the reason why the analysis of optimal taper focused only on how reduce training for peak performance. The computations were done for regular training and for a step 20% increase for 28 d before taper to remain within the range of the training effort actually produced by swimmers. Optimal training before taper cannot be searched because the gain in performance after optimal taper is linearly related to the training done before the reduction of training, despite limitations in training adaptations. In the models tested in the present study, only Model TI and Model TIF included inhibitory processes in an attempt to introduce limitations in the positive effect of training, and neither of them provided any improvement compared with Model Ba or Model Bu. Other modifications of Model Ba (38,39) have been proposed to take into consideration that positive effect is not simply proportional to TL. It was showed that modification of Model Ba with Hill saturation function resulted in better fit in elite swimmers analyzed over a period of 4 ± 2 yr (38). In line with the indirect-response models tested in this study, there are other formulations using delayed positive effect after training session (8,14,15,40). Using two exponentials for positive effect was found to outperform Model Bu with data sets from short-track sprint skaters (15). Extensions of Model Ba or Model Bu have been also formulated to account for previous training history (41) or temporal variation in model parameters (42,43). Methodologies other than impulse-response models also deserve attention: Per-Pot model (44), neural networks (45,46), exponentially weighed and rolling average models (16), and machine-learning models (4,5). One way to assess the relative strengths and limitations of these different models is to compare their performance using a common data set. This is why we believe that the database generated for the present study has considerable potential for reuse. It is rich enough to support testing of a wide range of models that predict performance from cumulated training. Nevertheless, the strongest limitation of these models is probably the use of one single metric for quantifying TL as model input (37). It does not allow us to take into consideration the specificities of the different types of exercises included in the training program. For example, variations in volume or intensity could give identical TL despite difference in their effects. Regardless of the variation in TL for peaking performance, how the content of sessions is determined, and how they are articulated over time, is a great part of the coach's work when planning training. More complex models or combination of different approaches are most likely needed to account for all the key factors in training planning.

CONCLUSIONS

In conclusion, this study showed that Model Ba and Model Bu were equally relevant for modeling the change in performance with training in athletes. Because of the lower number of adjustable parameters, Model Ba should be preferred to Model Bu to provide information about response to training. The interest in using Model Bu could arise from the inferences made from its application to dedicated large data sets, rather than precise prediction for a given individual. The observed accuracy of 2% to 3% for the prediction of one individual's performance in response to training was not satisfactory. The challenge to progress toward practical applications to assist coaches and athletes for planning training requires exploration of other approaches to modeling the responses to training.

The authors gratefully thank all the swimmers who participated in this study. The authors also acknowledge Crane Rogers for editing the manuscript for English language and helpful discussion.

The authors declare that they have no conflicting interests. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine.

APPLIED SCIENCES

REFERENCES

1. Rasche C, Pfeiffer M. Training. In: Baca A, Perl J, editors. *Modelling and Simulation in Sport and Exercise*. Routledge; 2018. pp. 187–207.
2. Stephens Hemingway B, Greig L, Jovanovic M, Ogorek B, Swinton P. Traditional and contemporary approaches to mathematical fitness-fatigue models in exercise science: a practical guide with resources. Part I. [Internet]. *SportRxiv*. 2021. Available from: osf.io/ap75j 2020.
3. Swinton P, Stephens Hemingway B, Rasche C, Pfeiffer M, Ogorek B. Traditional and contemporary approaches to mathematical fitness-fatigue models in exercise science: a practical guide with resources. Part II. [Internet]. *SportRxiv*. 2021. Available from: osf.io/5qgc2.
4. Imbach F, Perrey S, Chailan R, Meline T, Candau R. Training load responses modelling and model generalisation in elite sports. *Sci Rep*. 2022;12(1):1586.
5. Imbach F, Sutton-Charani N, Montmain J, Candau R, Perrey S. The use of fitness-fatigue models for sport performance modelling: conceptual issues and contributions from machine-learning. *Sports Med Open*. 2022;8(1):29.
6. Mujika I, Padilla S, Pyne D, Busso T. Physiological changes associated with the pre-event taper in athletes. *Sports Med*. 2004;34(13):891–927.
7. Banister EW, Calvert TW, Savage MV, Bach T. A systems model of training for athletic performance. *Aust J Sports Med*. 1975;7:57–61.
8. Calvert TW, Banister EW, Savage MV, Bach T. A systems model of the effects of training on physical performance. *IEEE Trans Syst Man Cybern*. 1976;6(2):94–102.
9. Stephens Hemingway BH, Burgess KE, Elyan E, Swinton PA. The effects of measurement error and testing frequency on the fitness-fatigue model applied to resistance training: a simulation approach. *Int J Sports Sci Coach*. 2020;15(1):60–71.
10. Stephens Hemingway B, Swinton P, Ogorek B. The suitability of a quasi-Newton algorithm for estimating fitness-fatigue models: sensitivity, troublesome local optima, and implications for future research (an in silico experimental design). [Internet]. *SportRxiv*. 2021. Available from: osf.io/dx7gm.
11. Vermeire K, Ghijs M, Bourgois JG, Boone J. The fitness–fatigue model: what's in the numbers? *Int J Sports Physiol Perform*. 2022; 17(5):810–3.
12. Busso T. Variable dose-response relationship between exercise training and performance. *Med Sci Sports Exerc*. 2003;35(7):1188–95.

Copyright © 2023 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

13. Busso T. From an indirect response pharmacodynamic model towards a secondary signal model of dose-response relationship between exercise training and physical performance. *Sci Rep*. 2017;7:40422.
14. Philippe AG, Borrani F, Sanchez AM, Py G, Candau R. Modelling performance and skeletal muscle adaptations with exponential growth functions during resistance training. *J Sports Sci*. 2019;37(3):254–61.
15. Borrani F, Solsona R, Candau R, Meline T, Sanchez AM. Modelling performance with exponential functions in elite short-track speed skaters. *J Sports Sci*. 2021;39(20):2378–85.
16. Mitchell LJG, Rattray B, Fowlie J, Saunders PU, Pyne DB. The impact of different training load quantification and modelling methodologies on performance predictions in elite swimmers. *Eur J Sport Sci*. 2020; 20(10):1329–38.
17. Burnham KP, Anderson DR. *Model Selection and Multimodel Inference*. 2nd ed. New York: Springer; 2002. p. 488.
18. Busso T, Carasso C, Lacour JR. Adequacy of a systems structure in the modeling of training effects on performance. *J Appl Physiol (1985)*. 1991;71(5):2044–9.
19. Spiess AN, Neumeyer N. An evaluation of R^2 as an inadequate measure for nonlinear models in pharmacological and biochemical research: a Monte Carlo approach. *BMC Pharmacol*. 2010;10:6.
20. Chalencon S, Pichot V, Roche F, et al. Modeling of performance and ANS activity for predicting future responses to training. *Eur J Appl Physiol*. 2015;115(3):589–96.
21. Mujika I, Busso T, Lacoste L, Barale F, Geyssant A, Chatard JC. Modeled responses to training and taper in competitive swimmers. *Med Sci Sports Exerc*. 1996;28(2):251–8.
22. R Core Team. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing; 2021. URL https://www.R-project.org/.
23. Thomas L, Mujika I, Busso T. A model study of optimal training reduction during pre-event taper in elite swimmers. *J Sports Sci*. 2008; 26(6):643–52.
24. Sanchez AM, Galbes O, Fabre-Guery F, et al. Modelling training response in elite female gymnasts and optimal strategies of overload training and taper. *J Sports Sci*. 2013;31(14):1510–9.
25. Meline T, Mathieu L, Borrani F, Candau R, Sanchez AM. Systems model and individual simulations of training strategies in elite short-track speed skaters. *J Sports Sci*. 2019;37(3):347–55.
26. Hopkins WG, Schabort EJ, Hawley JA. Reliability of power in physical performance tests. *Sports Med*. 2001;31(3):211–34.
27. Meeusen R, Duclos M, Foster C, et al. Prevention, diagnosis, and treatment of the overtraining syndrome: joint consensus statement of the European College of Sport Science and the American College of Sports Medicine. *Med Sci Sports Exerc*. 2013;45(1):186–205.
28. Pfeiffer M. Modeling the relationship between training and performance—a comparison of two antagonistic concepts. *Int J Comput Sci Sport*. 2008;7(2):13–32.
29. Clarke DC, Skiba PF. Rationale and resources for teaching the mathematical modeling of athletic training and performance. *Adv Physiol Educ*. 2013;37(2):134–52.
30. Morton RH. The quantitative periodization of athletic training: a model study. *Sports Med Training Rehabilitation*. 1991;3:19–28.
31. Fitz-Clarke JR, Morton RH, Banister EW. Optimizing athletic performance by influence curves. *J Appl Physiol (1985)*. 1991;71(3):1151–8.
32. Skorski S, Mujika I, Bosquet L, Meeusen R, Coutts AJ, Meyer T. The temporal relationship between exercise, recovery processes, and changes in performance. *Int J Sports Physiol Perform*. 2019;14(8):1015–21.
33. Thomas L, Busso T. A theoretical study of taper characteristics to optimize performance. *Med Sci Sports Exerc*. 2005;37(9):1615–21.
34. Thomas L, Mujika I, Busso T. Computer simulations assessing the potential performance benefit of a final increase in training during pre-event taper. *J Strength Cond Res*. 2009;23(6):1729–36.
35. Mujika I, Padilla S. Scientific bases for precompetition tapering strategies. *Med Sci Sports Exerc*. 2003;35(7):1182–7.
36. Bosquet L, Montpetit J, Arvisais D, Mujika I. Effects of tapering on performance: a meta-analysis. *Med Sci Sports Exerc*. 2007;39(8): 1358–65.
37. Busso T, Thomas L. Using mathematical modeling in training planning. *Int J Sports Physiol Perform*. 2006;1(4):400–5.
38. Hellard P, Avalos M, Millet G, Lacoste L, Barale F, Chatard JC. Modeling the residual effects and threshold saturation of training: a case study of Olympic swimmers. *J Strength Cond Res*. 2005;19(1):67–75.
39. Turner JD, Mazzoleni MJ, Little JA, Sequeira D, Mann BP. A nonlinear model for the characterization and optimization of athletic training and performance. *Biomed Hum Kinet*. 2017;9:82–93.
40. Matabuena M, Rodriguez-Lopez R. An improved version of the classical Banister model to predict changes in physical condition. *Bull Math Biol*. 2019;81(6):1867–84.
41. Ludwig M, Asteroth A, Rasche C, Pfeiffer M. Including the past: performance modeling using a preload concept by means of the fitness-fatigue model. *Int J Comput Sci Sport*. 2019;18(1):115–34.
42. Busso T, Denis C, Bonnefoy R, Geyssant A, Lacour JR. Modeling of adaptations to physical training by using a recursive least squares algorithm. *J Appl Physiol (1985)*. 1997;82(5):1685–93.
43. Kolossa D, Bin Azhar MA, Rasche C, et al. Performance estimation using the fitness-fatigue model with Kalman filter feedback. *Int J Comput Sci Sport*. 2016;16(2):117–29.
44. Perl J. PerPot: a metamodel for simulation of load performance interaction. *Eur J Sport Sci*. 2001;1(2):1–13.
45. Edelmann-nusser J, Hohmann A, Henneberg B. Modeling and prediction of competitive performance in swimming upon neural networks. *Eur J Sport Sci*. 2002;2(2):1–10.
46. Carrard J, Kloucek P, Gojanovic B. Modelling training adaptation in swimming using artificial neural network geometric optimisation. *Sports (Basel)*. 2020;8(1):8.

Copyright © 2023 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.